Hritwik Biyani

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EDUCATION

Stony Brook University — Stony Brook, New York
Master of Science in Computer Science (GPA: 4.0/4.0) — *August 2025 – January 2027*

MIT World Peace University — Pune, India
Bachelor of Technology in Computer Science — *July 2017 – July 2021*

EXPERIENCE

Software Engineer Intern — June 2026 – August 2026
Intuit — *New York, NY*

- Implemented inline task editing with a publish-subscribe system to guard unsaved changes across multiple pages; deployed to approximately 1.9M users
- Built reusable data pipelines and observability to support optimization of a LightGBM quick-action recommendation model on the QuickBooks home page, reducing p95 latency from approximately 640ms to 140ms
- Developed a real-time notification tray and email notification service for task management in Python using AWS DynamoDB, SQS, EventBridge, and Numaflow; served approximately 300K users in a pilot

Senior Data Engineer — July 2023 – July 2025
ZS Associates — *Pune, India*

- Engineered a fault-tolerant ingestion service integrating 40+ sources across REST APIs, FTP, and SQL, with structured error handling and retry logic; reduced time to production for new data sources by 4–5 weeks
- Designed an event-driven data governance system using serverless functions and asynchronous triggers, with approval workflows and audit logging for data change requests; enabled approximately 12x growth in data ingested
- Built agentic workflows for an account management service to research and populate customer-account news insights and SWOT analyses, using user-specific context management and dynamic prompts

Data Engineer — July 2021 – June 2023
ZS Associates — *Pune, India*

- Built a Node.js backend for real-time contracting scenario analysis, processing tens of thousands of rows with configurable algorithms to optimize contract selection and increase ROI by 30%+
- Developed a lazy-loaded graph visualization tool using D3.js and Neo4j, with optimized Cypher queries to render complex health system hierarchies through traversals across 10,000+ nodes

PROJECTS

Graduate Research Assistant | *Stony Brook University* — February 2026 – Present

- Implemented and formally verified the Stellar Consensus Protocol for a decentralized blockchain network using DistAlgo, a Python-based research language for distributed systems
- Developed a degree-planning constraint solver in Python using OR-Tools and DOcplex, with automated testing and benchmarking to compare against Prolog and clingo approaches for research analysis

SBUnix Operating System Kernel | *Stony Brook University* — February 2026 – Present

- Built an operating system kernel from scratch in C and Assembly, contributing file systems, terminal functionality, and memory allocators
- Developed a custom MicroPython port for the operating system

Sharded Transactional Database (2PC + Paxos) | *Stony Brook University* | Github repo — October 2025 – December 2025

- Built a distributed key-value store in Java supporting cross-shard ACID transactions and linearizable consistency
- Implemented Multi-Paxos replication with cluster reconfiguration and two-phase commit for cross-shard transactions
- Achieved 5,000+ transactions per second under concurrent workloads through iterative optimization guided by logging and monitoring

TECHNICAL SKILLS

Programming Languages: Python, Java, TypeScript, JavaScript, C, Bash, GraphQL
Skills: Data Structures, CI/CD, API Design, Abstraction, Operating Systems, Observability, Agile Principles, Reliable Systems
Tools / Frameworks: Git, React, AWS, Kafka, Spring Boot, Snowflake, Node.js, Flask, Docker, PostgreSQL, MySQL
AI skills: MCP Server, Prompt Engineering, AI Workflows, LangChain, ollama, Harness Engineering